Name: ____________________ **Worksheet**

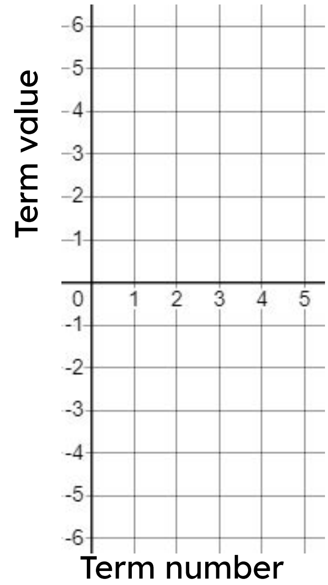
Representing sequences graphically

Task A: Plotting arithmetic sequences

1) Find and plot the first five terms of these arithmetic sequences.

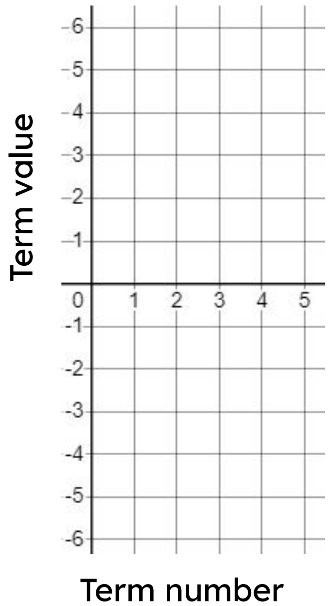
a) $3n - 9$

Term number (n)	1	2	3	4	5
Term value (T)	-6	-3	0	3	6

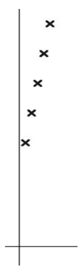


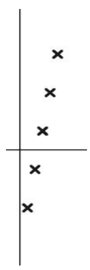
b) $3 - n$

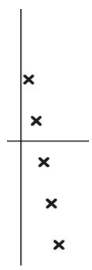
Term number (n)	1	2	3	4	5
Term value (T)	2	1	0	-1	-2



2) Match the sequences to their graph. $4n + 10$ $4n - 10$ $10 - 4n$



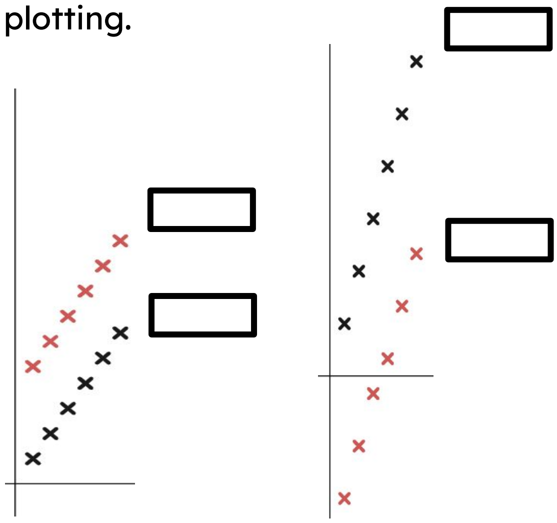




Task B: Features of sequences represented graphically

1) a) Match these arithmetic sequences to the correct plotting.

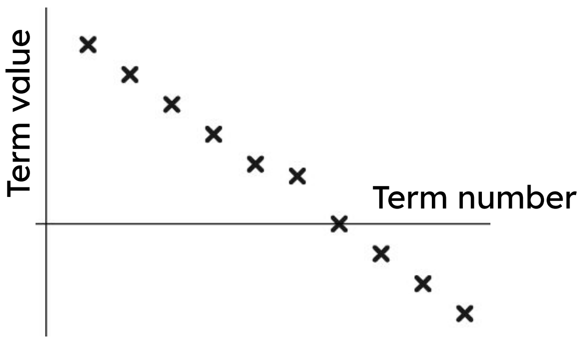
$3n \qquad \frac{3}{2}n \qquad \frac{3}{2}n + \frac{11}{2} \qquad 3n - 10$



b) Write a few sentences explaining how you identified each sequence.

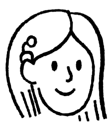
2) Izzy has plotted the sequence $5 - \frac{5}{7}n$

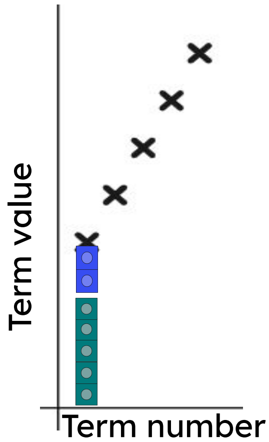
Which term number is wrong? Write a sentence to justify your answer.



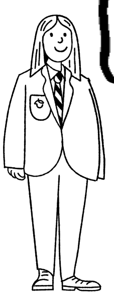
3) a) Sofia has started to draw the structure of the arithmetic sequence $2n + 5$
Finish her diagram for her.

b) Write a sentence explaining why this structure represents the sequence $2n + 5$





4) The 10^{th} term of the arithmetic sequence $3.5n$ is 35 and the 20^{th} term is 70 so Laura says…



I know the 10^{th} term of the arithmetic sequence $3.5n - 21$ is 14 so the 20^{th} term must be double that. It's 28

Explain to Laura why this does work for the arithmetic sequence $3.5n$ but it doesn't for the sequence $3.5n - 21$